



MIT SLOAN SCHOOL OF MANAGEMENT
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Glossary of frameworks

A

- **Agentic loop:** The construction-side description of tool use: model and tools exchanging messages in the background, sometimes for minutes or hours, before a final response is produced.
- **AgentOps and continuous evaluation:** The shift from CI/CD/CT/CM to a deployment discipline that treats continuous evaluation and skew detection as first-class concerns. AgentOps is the discipline; the dashboard is its instrument.
- **AI bill of materials:** Structured metadata disclosures for agentic systems, training data provenance, model architecture, version history, third-party components, and tool dependencies. Enables transparent governance and reduces regulatory friction.
- **AI continuum:** Classical machine learning → deep learning → generative AI → Agentic AI → AGI. Positions Agentic AI as a qualitative threshold on the continuum, not merely a quantitative step.
- **AI-first vs. AI-native:** AI-first organizations prioritize AI in their plans. AI-native organizations think and act with AI as the default. The second is the harder destination, and the one Agentic AI requires.
- **Autonomy spectrum (dial):** Autonomy as a calibrated dial, from fully constrained (pre-verified responses only) to fully autonomous (novel outputs and external actions). Most deployments should sit deliberately between, calibrated by context, stakes, and safeguard design.

B

- **Build-buy-stack:** The strategic posture for adoption at each layer of the stack: model, framework, platform. Decision should be driven by required autonomy and skill set, not by technology preference. Often a combination.

C

- **Comparison that matters:** Evaluate agent risk against the current process, not against perfection. Agent-with-safeguards versus the human-driven baseline, on the dimensions that matter.

E

- **Everyone is an agent manager:** The reframing of the leadership role: Each professional now directs a small team of agents alongside human colleagues; domain experts who supervise the agents that act in their domain.

F

- **Five building blocks of an agent:** Perception, action, planning, memory, and safety. Every agent design is a set of commitments on these five axes. Andreas's architectural spine for the program.
- **Four safeguard layers:** Hard-coded constraints, output verification, reversibility by design, escalation architecture. Each layer is independent of the others. Defense in depth.

G

- **G5 and intent over code:** Tim Kraska's research framing of the fifth-generation programming language: natural language and a system ontology as the new source code, with all other code derived automatically. The horizon of how organizations build and operate software.
- **Governance as differentiator:** Organizations that invest in transparent, auditable governance signal trustworthiness to markets, customers, and regulators, and reduce regulatory scrutiny. Not a compliance cost; a competitive position.

H

- **Human-in-the-loop with thresholds:** Autonomy is calibrated by stakes. The design pattern is to set thresholds: At one level the agent acts; above another it escalates; above another a human originates the decision.

L

- **Layered lines of defense:** Data and model governance as the foundation. Runtime input/output guardrails as the operational layer. Structured metadata disclosures (the AI bill of materials) as the transparency layer.

M

- **Methodology-as-workflow:** If you can articulate a methodology, Porter's Five Forces, an account-enrichment process, you can encode it as an agentic workflow. The conceptual bridge between business process and software.

N

- **Non-human identity management:** Agents have identities, permissions, and access rights. They must be managed with disciplines built over decades for human users, least-privilege access, defined scope, periodic review, and revocation.

R

- **Reactive vs. proactive:** The program's central distinction. Reactive systems take a prompt and produce a response. Proactive systems take a goal and produce a plan, an execution, and an adaptation. The expert witness vs. the project manager.

S

- **Suitcase words:** Marvin Minsky's term for words we pack with multiple meanings and carry around as if they were one thing: agent, intelligence, autonomy. Words shared in use, not in meaning, are organizational hazards.
- **Scaling paradox:** Scaling is not enlarging a pilot. It is transitioning from a controlled environment to an operational context that no one has fully tested. The failure modes change; the governance requirements intensify.
- **Start small, try a lot:** As a deployment strategy: not ambitious monolithic initiatives, but a portfolio of focused experiments. Bottom-up empowerment is combined with selective top-down strategic initiatives.

T

- **Three adoption realities:** (1) Speed of iteration matters more than perfection of plan. (2) The biggest barriers are organizational, not technical. (3) Existing metrics will not capture the value agentic systems create.
- **Three weeks to MVP:** From design board to first internal release (fishfooding) in three weeks. The discipline is held by holding the scope tight, not the timeline loose.